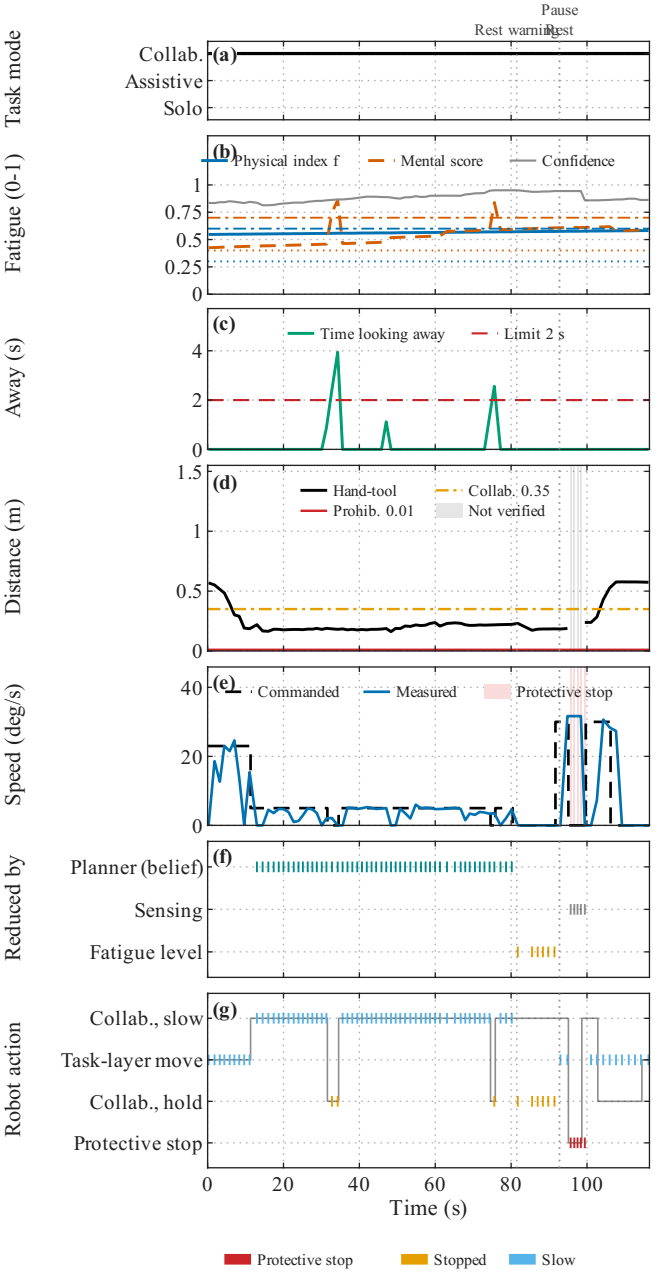

Pause
Rest warning
Rest
Task mode
Collab.
Assistive
Solo
(a)
Fatigue (0-1)
(b)
Physical index f
Mental score
Confidence
1
0.75
0.5
0.25
0
Away (s)
(c)
Time looking away
Limit 2 s
4
2
0
Distance (m)
(d)
Hand-tool
Collab. 0.35
Prohib. 0.01
Not verified
1.5
1
0.5
0
Speed (deg/s)
(e)
Commanded
Measured
Protective stop
40
20
0
Reduced by
(f)
Planner (belief)
Sensing
Fatigue level
Robot action
(g)
Collab., slow
Task-layer move
Collab., hold
Protective stop
0
20
40
60
80
100
Time (s)
Protective stop
Stopped
Slow
Planner (nominal)